

Low-Cost DUSK-DAWN CONTROLLER

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Solar streetlights can be easily integrated with a dusk-dawn controller by simply employing a pnp transistor and a few resistors where the solar panel itself works as the sensor. But what about other lighting sources that do not employ solar panels such as automatic lighting systems in small wind turbines, automatic lighting in cars or battery based systems where automatic lighting is necessary?

A simple, low-cost, yet an effective solution for the dusk-dawn controller circuit is described here. Author's prototype is shown in Fig. 1.

Circuit and working

Circuit diagram of the dusk-

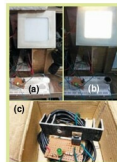


Fig. 1: Author's prototype: (a) dawn mode, (b) dusk mode and (c) prototype

dawn controller is shown in Fig. 2. It is built around a light-dependent resistor (LDR1), n-channel MOSFET IRF640 (T1), 12V LED light or a small inverter (100W) and a few other components.

The 12V battery-operated circuit is designed such that the common battery supply is used for operating the circuit as well as for load, that

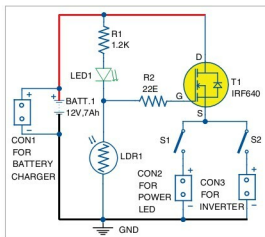


Fig. 2: Circuit diagram of the dusk-dawn controller

is, for power LED/small inverter circuit. Resistors R1 and R2 are used as a voltage divider and a current limiter in the circuit, respectively. LED1 is used as circuit de-activation indicator. LDR1 is the main com-

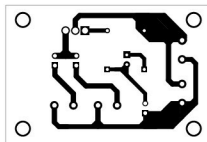


Fig. 3: Actual-size PCB pattern of the dusk-dawn controller

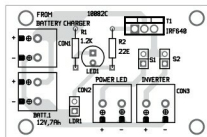


Fig. 4: Component layout of the PCB

ponent for actuation of the dusk-to-dawn sensing. The n-channel MOSFET IRF640 is for the switching action of the LED light or the small inverter connected to the system through switches S1 and S2, respectively.

With daylight falling on LDR1 (dawn mode), resistance of LDR1 becomes low, a small current flows through LED1 and it glows. At the same time, due to low voltage across LDR1, MOSFET IRF640 does not trigger. So load (LED light/inverter) remains off.

But when the surrounding area of LDR1 (dusk mode) is dark (at night), resistance of LDR1 becomes very high, no current flows through LED1 and it does not glow. At the same time, due to higher voltage across LDR1, IRF640 triggers. So load light (power LED light/ inverter) comes on, provided switch S1 or S2 is on.

IRF640 can handle a maximum current of approximately 18A, but you can limit its application to approximately 8A-9A only, based on which the heat-sink is attached to the MOSFET.

Construction and testing

An actual-size, single-side PCB for the dusk-dawn controller is shown in Fig. 3 and its component layout in Fig. 4.

After assembling the circuit, enclose it in a suitable box. The unit should be fixed on the pillar in such a way that the daylight falls directly on LDR1. **EFY**



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PARTS LIST

Semiconductors:

LED1

T1 = IRE640, n=

Basistone, cell 1/4 south, +5% carbon.

Resistors (all 1/4-watt, $\pm 5\%$ carbon):
R1 1.2 k Ω , chm

R1 - 1.2-kilo-ohm
R2 - 22-ohm

K2 - 22-0111

Miscellaneous:

LDRI	- Epoxy-coated, moisture-resistant 25mm/12mm diameter, light-dependent resistor
S1-S2	- On/off switch
BATT.1	- 12V, 7Ah battery
CON1-CON3	- 2-pin connector terminal
	- 12V DC, 5W LED lights
	- 100W small inverter

S1-S2

BATT 1 = 12V 7Ah battery

CON1-CON3 - 2-pin connector terminal

- 12V DC, 5W LED light

- 100W small inverter